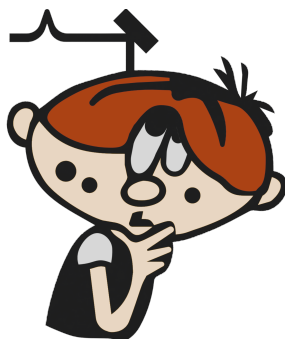

LCS RH/T Lab staton



User's manual

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1 Introduction

LCS RH/T Lab staton is device that allows measuring *relative humidity* (RH) and *temperature*. Device interfaces with LCS' Raspberry Pi server using WIFI and MQTT.

1.1 Description

The device is composed of the following parts (as pictured on pic. [1](#)):

1. Central device (ESP32-S3 Zero Board)
2. RH/T sensor (Sensirion SHT35) + protective casing
3. FPC/RJ45 adapter cable



Figure 1: Disassembled device. 1 - main housing, 2 - status LED, 3/4 - RH/T sensor cover upper/lower, 5 - sensor cable



Figure 2: Main housing (left - RJ45, right - USB-C)

1.2 Power

Device requires 5V provided via USB-C. Due to constant operation as WIFI-STA it requires too much energy for extended battery operations. Connecting it via phone charger is recommended but the standard PC USB port is sufficient.

2 Installation

1. Plug the usb-C cable with power into the central device
2. When the LED lights up red the device is powered on and is looking for the sensor presense
3. Instert sensor into the FPC side of the interface cable (as pictured on [3](#)) and install the protective casing
4. Insert the cable into the RJ45 recepticle. If sensor is detected the LED will stop being red.
5. The LED should now turn blue, indicating that the station attempts to connect to WIFI and MQTT.
6. Once the LED turns off the devices is connected and sends data each second to MQTT (seperate channel for graphing that is refreshed every 5 seconds).

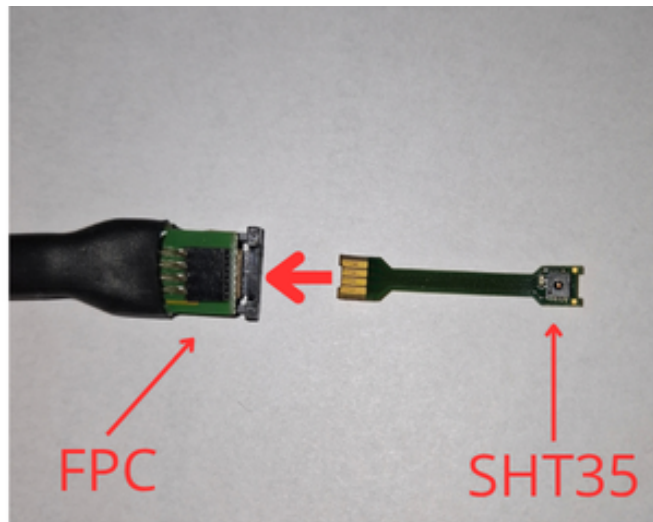


Figure 3: SHT35 sensor

3 Getting data

Data is available via MQTT. The broker URI is `mqtt://1cs.ktp.pwr.edu.pl:1883`. Relevant topics are:

- `lab/readings` (1s period measurement data)
- `lab/RHT_graph` (5s period data for graphing)

Data can be saved using the dedicated logger on the *Laboratory 142* tab, or appended to the specific chamber logs.

4 Development options

By plugging the station into a PC and running a serial monitor the logs are available. To change the the WIFI credential or MQTT URI edit the `secrets.h` file. Do not upload the file into the git.